

EXPERIMENT 3

Water Jug Problem

Question

Write a Python program to solve the Water Jug problem .

Aim

To write a Python program to solve the **Water Jug problem** using the **Breadth First Search (BFS)** algorithm.

Algorithm

1. Start with two empty water jugs.
2. Set the capacities of the jugs as 4 litres and 3 litres.
3. Define the goal as obtaining 2 litres of water.
4. Generate possible operations:
 - Fill the 4-litre jug.
 - Fill the 3-litre jug.
 - Empty the 4-litre jug.
 - Empty the 3-litre jug.
 - Transfer water from one jug to another.
5. Store each new state in a queue.
6. Check whether the required 2 litres is obtained.
7. Continue until the goal state is reached.
8. Display the solution.

Code

```
python exp3.py - C:/Users/royal/python exp3.py (3.13.14)
File Edit Format Run Options Window Help
from collections import deque

def water_jug():
    capacity1 = 4
    capacity2 = 3
    target = 2

    queue = deque()
    queue.append((0, 0))

    visited = set()
    visited.add((0, 0))

    while queue:
        jug1, jug2 = queue.popleft()

        print(jug1, jug2)

        if jug1 == target or jug2 == target:
            print("Goal reached")
            return

        states = [
            (capacity1, jug2),
            (jug1, capacity2),
            (0, jug2),
            (jug1, 0),
            (jug1 - min(jug1, capacity2 - jug2),
             jug2 + min(jug1, capacity2 - jug2)),
            (jug1 + min(jug2, capacity1 - jug1),
             jug2 - min(jug2, capacity1 - jug1))
        ]

        for state in states:
            if state not in visited:
                visited.add(state)
                queue.append(state)

water_jug()
```

Output

```
===== RESTART: C:/Users/royal/python exp3.py =  
0 0  
4 0  
0 3  
4 3  
1 3  
3 0  
1 0  
3 3  
0 1  
4 2  
Goal reached  
|
```

Result

Thus, the **Water Jug problem was successfully solved using the Breadth First Search algorithm**, and the required **2 litres of water was obtained**.